Building quantum codes with transversal CCZ

from projective geometry

Bohan Lu, Duke University

Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

Quantum error correction

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. The Pauli matrices $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $Y = iXZ$ and I span the 2×2 complex matrices, so an error on one qubit expands in I, X, Y, Z. The weight of a Pauli product on n qubits is the number of qubits where it is not I.

[Got97] D. Gottesman, Stabilizer codes and quantum error correction, Ph.D. thesis, Caltech (1997), arXiv:quant-ph/9705052.

Quantum error correction

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. The Pauli matrices $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $Y = iXZ$ and I span the 2×2 complex matrices, so an error on one qubit expands in I, X, Y, Z. The weight of a Pauli product on n qubits is the number of qubits where it is not I.

- A *stabilizer code* is the joint $+1$ eigenspace of an abelian group $\mathcal{S}$ of Pauli products with $-I \notin \mathcal{S}$ [Got97]. Its generators are the checks.
- Measuring the checks gives the syndrome. An error flips the outcome of every check it anticommutes with.

[Got97] D. Gottesman, Stabilizer codes and quantum error correction, Ph.D. thesis, Caltech (1997), arXiv:quant-ph/9705052.

Quantum error correction

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. The Pauli matrices $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $Y = iXZ$ and I span the 2×2 complex matrices, so an error on one qubit expands in I, X, Y, Z. The weight of a Pauli product on n qubits is the number of qubits where it is not I.

- A *stabilizer code* is the joint $+1$ eigenspace of an abelian group $\mathcal{S}$ of Pauli products with $-I \notin \mathcal{S}$ [Got97]. Its generators are the checks.
- Measuring the checks gives the syndrome. An error flips the outcome of every check it anticommutes with.
- $[\![n, k, d]\!]$: n qubits, a code space of dimension 2^k, and every Pauli product of weight less than d is detected or acts as a scalar on the code space.

[Got97] D. Gottesman, Stabilizer codes and quantum error correction, Ph.D. thesis, Caltech (1997), arXiv:quant-ph/9705052.

CSS codes: two binary codes

X_u flips bit u of $|c\rangle$, and $Z_u|c\rangle = (-1)^{c_u}|c\rangle$. For $v, z \in \mathbb{F}_2^n$, $X^v = \prod_u X_u^{v_u}$ and $Z^z = \prod_u Z_u^{z_u}$ satisfy $X^v Z^z = (-1)^{v\cdot z} Z^z X^v$.

For binary codes $\mathcal{C}_2 \subset \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] has the checks X^v, $v \in \mathcal{C}_2$, and Z^z, $z \in \mathcal{C}_1^\perp$. They commute because $\mathcal{C}_2 \perp \mathcal{C}_1^\perp$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes: two binary codes

X_u flips bit u of $|c\rangle$, and $Z_u|c\rangle = (-1)^{c_u}|c\rangle$. For $v, z \in \mathbb{F}_2^n$, $X^v = \prod_u X_u^{v_u}$ and $Z^z = \prod_u Z_u^{z_u}$ satisfy $X^v Z^z = (-1)^{v \cdot z} Z^z X^v$.

For binary codes $\mathcal{C}_2 \subset \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] has the checks X^v, $v \in \mathcal{C}_2$, and Z^z, $z \in \mathcal{C}_1^\perp$. They commute because $\mathcal{C}_2 \perp \mathcal{C}_1^\perp$.

- X^e flips Z^z when $z \cdot e = 1$. It is undetected if and only if $e \in \mathcal{C}_1$, so correcting X errors is decoding $\mathcal{C}_1$.
- Z^e is undetected if and only if $e \in \mathcal{C}_2^\perp$, so correcting Z errors is decoding $\mathcal{C}_2^\perp$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes: two binary codes

X_u flips bit u of $|c\rangle$, and $Z_u|c\rangle = (-1)^{c_u}|c\rangle$. For $v, z \in \mathbb{F}_2^n$, $X^v = \prod_u X_u^{v_u}$ and $Z^z = \prod_u Z_u^{z_u}$ satisfy $X^v Z^z = (-1)^{v\cdot z} Z^z X^v$.

For binary codes $\mathcal{C}_2 \subset \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] has the checks X^v, $v \in \mathcal{C}_2$, and Z^z, $z \in \mathcal{C}_1^\perp$. They commute because $\mathcal{C}_2 \perp \mathcal{C}_1^\perp$.

- X^e flips Z^z when $z \cdot e = 1$. It is undetected if and only if $e \in \mathcal{C}_1$, so correcting X errors is decoding $\mathcal{C}_1$.
- Z^e is undetected if and only if $e \in \mathcal{C}_2^\perp$, so correcting Z errors is decoding $\mathcal{C}_2^\perp$.
- Logical operators are X^v, $v \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and Z^z, $z \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$. So $k = \dim \mathcal{C}_1 - \dim \mathcal{C}_2$, $d = \min(d_X, d_Z)$ with $d_X = \min \mathrm{wt}(\mathcal{C}_1 \setminus \mathcal{C}_2)$ and $d_Z = \min \mathrm{wt}(\mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp)$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes: two binary codes

X_u flips bit u of $|c\rangle$, and $Z_u|c\rangle = (-1)^{c_u}|c\rangle$. For $v, z \in \mathbb{F}_2^n$, $X^v = \prod_u X_u^{v_u}$ and $Z^z = \prod_u Z_u^{z_u}$ satisfy $X^v Z^z = (-1)^{v\cdot z} Z^z X^v$.

For binary codes $\mathcal{C}_2 \subset \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] has the checks X^v, $v \in \mathcal{C}_2$, and Z^z, $z \in \mathcal{C}_1^\perp$. They commute because $\mathcal{C}_2 \perp \mathcal{C}_1^\perp$.

- X^e flips Z^z when $z \cdot e = 1$. It is undetected if and only if $e \in \mathcal{C}_1$, so correcting X errors is decoding $\mathcal{C}_1$.
- Z^e is undetected if and only if $e \in \mathcal{C}_2^\perp$, so correcting Z errors is decoding $\mathcal{C}_2^\perp$.
- Logical operators are X^v, $v \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and Z^z, $z \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$. So $k = \dim \mathcal{C}_1 - \dim \mathcal{C}_2$, $d = \min(d_X, d_Z)$ with $d_X = \min \mathrm{wt}(\mathcal{C}_1 \setminus \mathcal{C}_2)$ and $d_Z = \min \mathrm{wt}(\mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp)$.

With $\mathcal{C}_2 = \mathrm{rowspan}\, S$ and $\mathcal{C}_1 = \mathrm{rowspan}\, \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$, the logical basis is $|x\rangle_L = |\mathcal{C}_2|^{-1/2} \sum_{v\in\mathcal{C}_2} |xK \oplus v\rangle$, $x \in \mathbb{F}_2^k$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

Transversal gates and their limits

A logical gate $\bar{U}$ is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space. A fault in one U_j remains a single-qubit error.

$$H = \tfrac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}), \quad \mathrm{CCZ}|x_1 x_2 x_3\rangle = (-1)^{x_1 x_2 x_3}|x_1 x_2 x_3\rangle.$$

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and their limits

A logical gate $\bar{U}$ is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space. A fault in one U_j remains a single-qubit error.

$$H = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix} \right), \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}), \quad \mathrm{CCZ}|x_1 x_2 x_3\rangle = (-1)^{x_1 x_2 x_3} |x_1 x_2 x_3\rangle.$$

- H, S and CNOT generate the Clifford group, which is not universal. Adding T, or adding CCZ, gives a universal gate set.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and their limits

A logical gate $\bar{U}$ is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space. A fault in one U_j remains a single-qubit error.

$$H = \tfrac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}), \quad \mathrm{CCZ}|x_1 x_2 x_3\rangle = (-1)^{x_1 x_2 x_3}|x_1 x_2 x_3\rangle.$$

- H, S and CNOT generate the Clifford group, which is not universal. Adding T, or adding CCZ, gives a universal gate set.
- Eastin–Knill [EK09]: if a code detects every single-qubit error, its transversal logical gates are not universal. Each code has a limited set of transversal gates.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and their limits

A logical gate $\bar{U}$ is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space. A fault in one U_j remains a single-qubit error.

$$H = \tfrac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}), \quad \mathrm{CCZ}|x_1 x_2 x_3\rangle = (-1)^{x_1 x_2 x_3}|x_1 x_2 x_3\rangle.$$

- H, S and CNOT generate the Clifford group, which is not universal. Adding T, or adding CCZ, gives a universal gate set.
- Eastin–Knill [EK09]: if a code detects every single-qubit error, its transversal logical gates are not universal. Each code has a limited set of transversal gates.
- Magic-state distillation [BK05] turns many noisy copies of a non-Clifford state into fewer accurate copies, using Clifford gates. Each copy implements one non-Clifford gate.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S, CNOT. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode $[15,4,8]$	$T^{\otimes 15}$ acts as logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	A layer of T and $T^\dagger$ acts as logical CCZ.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S, CNOT. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode $[15,4,8]$	$T^{\otimes 15}$ acts as logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	A layer of T and $T^\dagger$ acts as logical CCZ.

CNOT between two blocks is transversal for every CSS code. Since Clifford $+T$ is universal, [EK09] rules out the missing gates.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

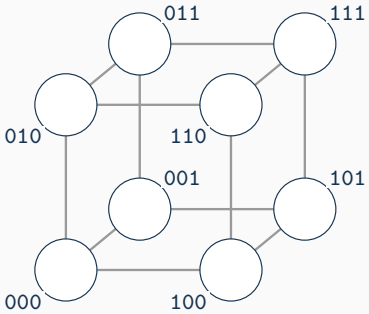
Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S, CNOT. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode $[15,4,8]$	$T^{\otimes 15}$ acts as logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	A layer of T and $T^\dagger$ acts as logical CCZ.

CNOT between two blocks is transversal for every CSS code. Since Clifford $+T$ is universal, [EK09] rules out the missing gates.

Question: which CSS codes with $k = 3$ have a choice $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ acting as logical CCZ?

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

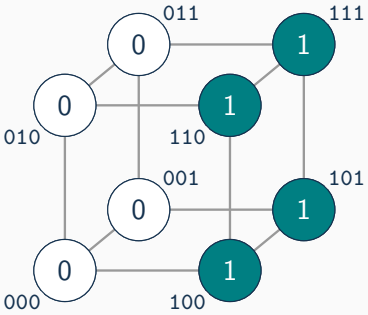
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

Qubits indexed by the points of $\mathbb{F}_2^3$



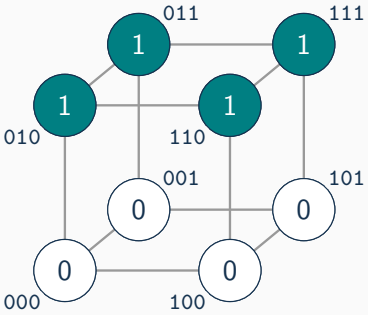
The vector u_1.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

Qubits indexed by the points of $\mathbb{F}_2^3$



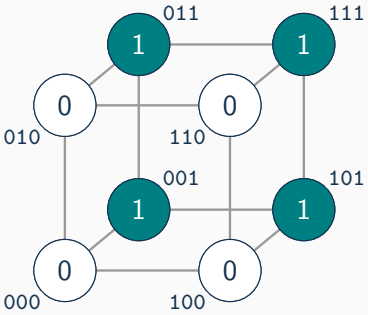
The vector u_2.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

Qubits indexed by the points of $\mathbb{F}_2^3$



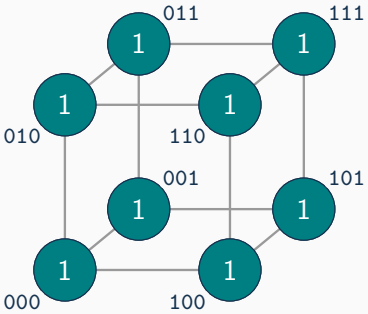
The vector u_3.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

Qubits indexed by the points of $\mathbb{F}_2^3$



The constant vector $\mathbf{1}$.

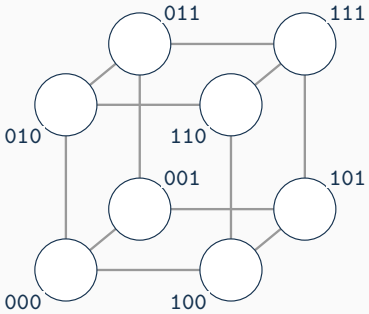
Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

$\mathrm{RM}(r, m)$ is the code of the vectors given by polynomials of degree at most r. For example, $\mathrm{RM}(1, 3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$ is an $[8, 4, 4]$ code.

The cube code $[\![8, 3, 2]\!]$

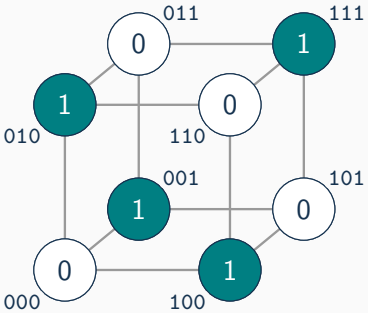


Take $S = \mathbf{1}$, the single check $X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$xK = x \cdot u = x_1u_1 \oplus x_2u_2 \oplus x_3u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

The cube code $[\![8, 3, 2]\!]$



The codeword $x \cdot u$ for $x = 111$.

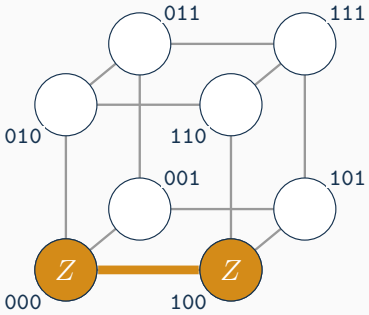
Take $S = \mathbf{1}$, the single check $X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$xK = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

For $x = 111$ the codeword $x \cdot u$ is $u_1 \oplus u_2 \oplus u_3$.

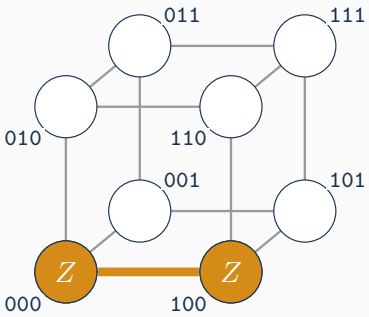
Distances of the cube code



The operator $Z_{000}Z_{100}$.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Then $Z_{000}Z_{100} = Z^z$ with $z = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

Distances of the cube code

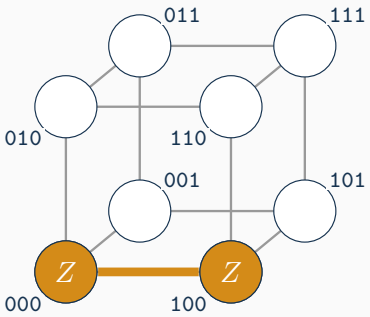


The operator $Z_{000}Z_{100}$.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Then $Z_{000}Z_{100} = Z^z$ with $z = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

- z has even overlap with $\mathbf{1}$, so Z^z commutes with the check $X^{\otimes 8}$.

Distances of the cube code

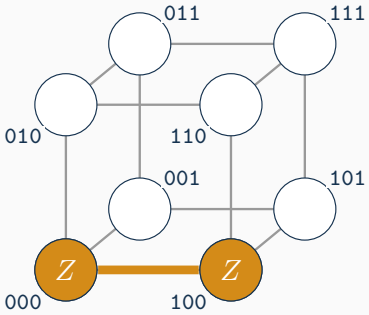


The operator $Z_{000}Z_{100}$.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Then $Z_{000}Z_{100} = Z^z$ with $z = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

- z has even overlap with $\mathbf{1}$, so Z^z commutes with the check $X^{\otimes 8}$.
- Every codeword is $c = b \oplus x \cdot u$, with $c(000) = b$ and $c(100) = b \oplus x_1$. So $Z^z|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.

Distances of the cube code

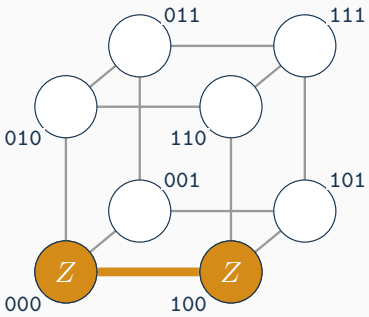


The operator $Z_{000}Z_{100}$.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Then $Z_{000}Z_{100} = Z^z$ with $z = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

- z has even overlap with $\mathbf{1}$, so Z^z commutes with the check $X^{\otimes 8}$.
- Every codeword is $c = b \oplus x \cdot u$, with $c(000) = b$ and $c(100) = b \oplus x_1$. So $Z^z|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.
- $\mathrm{RM}(1,3)$ has minimum weight 4, so $d_X = 4$.

Distances of the cube code



The operator $Z_{000}Z_{100}$.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Then $Z_{000}Z_{100} = Z^z$ with $z = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

- z has even overlap with $\mathbf{1}$, so Z^z commutes with the check $X^{\otimes 8}$.
- Every codeword is $c = b \oplus x \cdot u$, with $c(000) = b$ and $c(100) = b \oplus x_1$. So $Z^z|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.
- $\mathrm{RM}(1,3)$ has minimum weight 4, so $d_X = 4$.

This is the smallest interesting colour code: a transversal $T/T^\dagger$ layer gives logical CCZ [CH17].

Codeword weight, physical phase, and logical phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $W_\Gamma(c)$ counts the ones of c at T sites minus the ones at $T^\dagger$ sites.

Codeword weight, physical phase, and logical phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

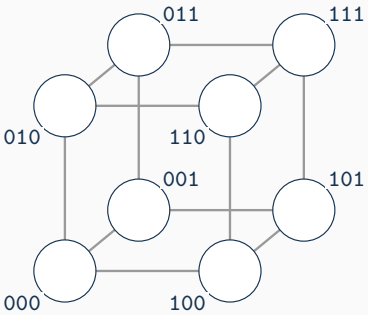
$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $W_\Gamma(c)$ counts the ones of c at T sites minus the ones at $T^\dagger$ sites.

The product acts as logical CCZ if and only if every codeword of the coset $xK \oplus \mathcal{C}_2$ receives the phase $(-1)^{x_1x_2x_3}$:

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

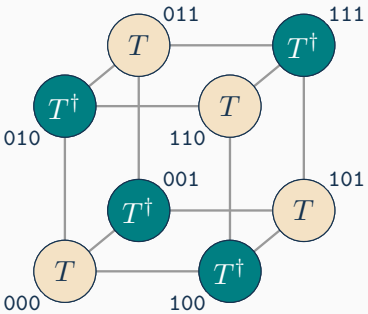
Choosing the signs on the cube



Every codeword of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so $f_{x,b}$ has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$. It is the codeword $x \cdot u$ for $x = 111$.

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

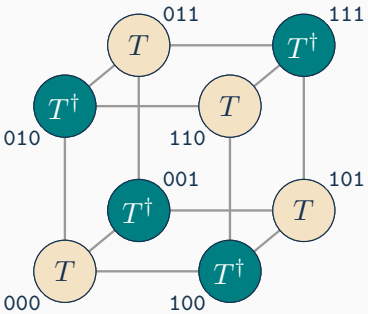
Every codeword of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so $f_{x,b}$ has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$. It is the codeword $x \cdot u$ for $x = 111$.

Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. With the Hamming distance $d_H(p, c) = \mathrm{wt}(p \oplus c)$,

$$W_\Gamma(c) = \mathrm{wt}(c) - 2\,\mathrm{wt}(p \wedge c) = d_H(p, c) - 4.$$

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

Every codeword of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so $f_{x,b}$ has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$. It is the codeword $x \cdot u$ for $x = 111$.

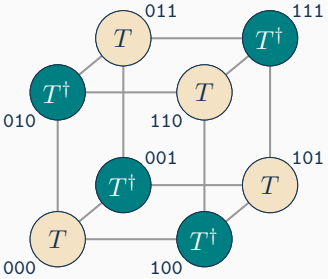
Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. With the Hamming distance $d_H(p, c) = \mathrm{wt}(p \oplus c)$,

$$W_\Gamma(c) = \mathrm{wt}(c) - 2\,\mathrm{wt}(p \wedge c) = d_H(p, c) - 4.$$

$x \neq 111$: $p \oplus f_{x,b} = b \oplus (x \oplus 111) \cdot u$ is nonconstant, so $d_H = 4$ and $W_\Gamma = 0$.

$x = 111$: $f_{x,b} \in \{p, \mathbf{1} \oplus p\}$, so $d_H \in \{0, 8\}$ and $W_\Gamma = \pm 4$, phase -1.

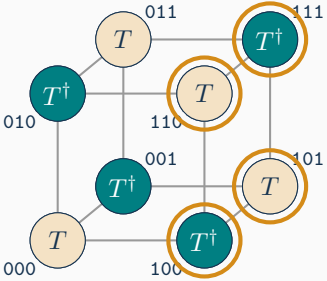
Reading the cube phase from overlaps



If k of the binary vectors $a_1, \ldots, a_m$ are 1 at u, then $\sum_{t=1}^{k} \binom{k}{t}(-2)^{t-1} = \frac{1-(-1)^k}{2} = (a_1 \oplus \cdots \oplus a_m)(u)$. Weighting u by Γ_u and summing gives $W_\Gamma(a_1 \oplus \cdots \oplus a_m) = \sum_{J \neq \emptyset} (-2)^{|J|-1} W_\Gamma(\bigwedge_{j \in J} a_j)$. For $c = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$:

$$W_\Gamma(c) = \sum_i x_i W_\Gamma(K_i) - 2 \sum_{i<j} x_i x_j W_\Gamma(K_i \wedge K_j) + 4 x_1 x_2 x_3 \, W_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

Reading the cube phase from overlaps



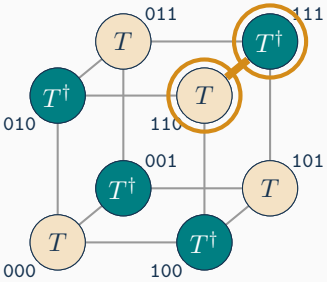
$K_1 = u_1$: the face $u_1 = 1$.

If k of the binary vectors $a_1, \dots, a_m$ are 1 at u, then $\sum_{t=1}^{k} \binom{k}{t}(-2)^{t-1} = \frac{1-(-1)^k}{2} = (a_1 \oplus \cdots \oplus a_m)(u)$. Weighting u by Γ_u and summing gives $W_\Gamma(a_1 \oplus \cdots \oplus a_m) = \sum_{J \neq \emptyset} (-2)^{|J|-1} W_\Gamma(\bigwedge_{j \in J} a_j)$. For $c = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$:

$$W_\Gamma(c) = \sum_i x_i W_\Gamma(K_i) - 2 \sum_{i<j} x_i x_j W_\Gamma(K_i \wedge K_j) + 4 x_1 x_2 x_3 \, W_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $W_\Gamma(K_1) = 0$. Likewise for K_2, K_3.

Reading the cube phase from overlaps



$K_1 \wedge K_2 = u_1 u_2$.

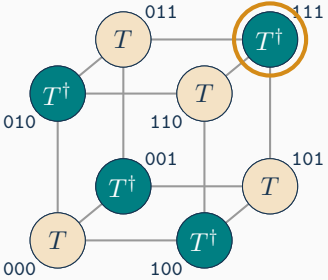
If k of the binary vectors $a_1, \dots, a_m$ are 1 at u, then $\sum_{t=1}^{k} \binom{k}{t}(-2)^{t-1} = \frac{1-(-1)^k}{2} = (a_1 \oplus \cdots \oplus a_m)(u)$. Weighting u by Γ_u and summing gives $W_\Gamma(a_1 \oplus \cdots \oplus a_m) = \sum_{J \neq \emptyset} (-2)^{|J|-1} W_\Gamma(\bigwedge_{j \in J} a_j)$. For $c = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$:

$$W_\Gamma(c) = \sum_i x_i W_\Gamma(K_i) - 2 \sum_{i<j} x_i x_j W_\Gamma(K_i \wedge K_j) + 4 x_1 x_2 x_3 \, W_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $W_\Gamma(K_1) = 0$. Likewise for K_2, K_3.

The edge $\{110, 111\}$ has one T and one $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2) = 0$. Likewise for the other pairs.

Reading the cube phase from overlaps



$K_1 \wedge K_2 \wedge K_3 = u_1 u_2 u_3$.

If k of the binary vectors $a_1, \ldots, a_m$ are 1 at u, then $\sum_{t=1}^{k} \binom{k}{t}(-2)^{t-1} = \frac{1-(-1)^k}{2} = (a_1 \oplus \cdots \oplus a_m)(u)$. Weighting u by Γ_u and summing gives $W_\Gamma(a_1 \oplus \cdots \oplus a_m) = \sum_{J \neq \emptyset} (-2)^{|J|-1} W_\Gamma(\bigwedge_{j \in J} a_j)$. For $c = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$:

$$W_\Gamma(c) = \sum_i x_i W_\Gamma(K_i) - 2 \sum_{i<j} x_i x_j W_\Gamma(K_i \wedge K_j) + 4 x_1 x_2 x_3 \, W_\Gamma(K_1 \wedge K_2 \wedge K_3).$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $W_\Gamma(K_1) = 0$. Likewise for K_2, K_3.

The edge $\{110, 111\}$ has one T and one $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2) = 0$. Likewise for the other pairs.

The vertex 111 is a $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2 \wedge K_3) = -1$ and $W_\Gamma(c) = -4 x_1 x_2 x_3 \equiv 4 x_1 x_2 x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with the overlap formula and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
$$W_\Gamma(g_a \wedge g_b) \equiv 0 \pmod 4,$$
$$W_\Gamma(g_a \wedge g_b \wedge g_c) \equiv [\{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}] \pmod 2.$$

Here $[P]$ is 1 if the statement P holds and 0 otherwise.

The gate condition becomes conditions on generator rows

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with the overlap formula and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
$$W_\Gamma(g_a \wedge g_b) \equiv 0 \pmod 4,$$
$$W_\Gamma(g_a \wedge g_b \wedge g_c) \equiv [\{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}] \pmod 2.$$

Here $[P]$ is 1 if the statement P holds and 0 otherwise.

The $[\![8, 3, 2]\!]$ code satisfies all three: rows and faces have $W_\Gamma = 0$, edges have $W_\Gamma = 0$, and $u_1u_2u_3$ has weight 1.

Historical context

2012 Bravyi–Haah [BH12]: triorthogonal matrices, transversal T up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via RM indicators.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows label logical T outputs. Even-weight rows give X checks.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j = 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every vector in $\mathcal{H}$ has even weight.

Nezami–Haah classify the parent spaces $\mathcal{H}$, then obtain codes by puncturing coordinates. This work fixes S from geometry first, then searches for a compatible K.

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17]. They give CCZ up to a diagonal Clifford correction (*quasi-transversal*).

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17]. They give CCZ up to a diagonal Clifford correction (*quasi-transversal*).

A physical $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$Q_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- The stabilizer code is fixed by projective geometry: $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$.
- The logical rows come from a SAT search.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- We found codes at block lengths 48 to 496 using S constructed from $\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2)$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. Every codeword of $\mathcal{C}_2$ has weight $\equiv 0 \pmod 8$. By the overlap identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. Every codeword of $\mathcal{C}_2$ has weight $\equiv 0 \pmod 8$. By the overlap identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.

Fix S using finite/projective geometry code

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. Every codeword of $\mathcal{C}_2$ has weight $\equiv 0 \pmod 8$. By the overlap identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.
4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \implies \text{weight-one errors are detected}, \qquad \beta_i \neq \beta_j \implies \beta_i \oplus \beta_j \neq 0.$$

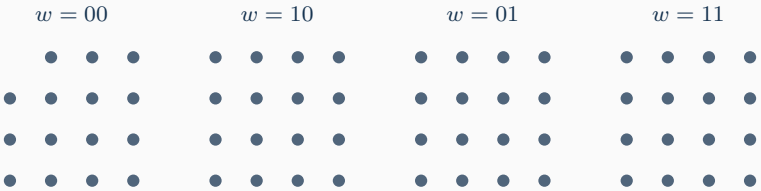
Distinct nonzero columns are points of $\mathrm{PG}(s-1,2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the stabilizer columns, independent of the choice of logical rows.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

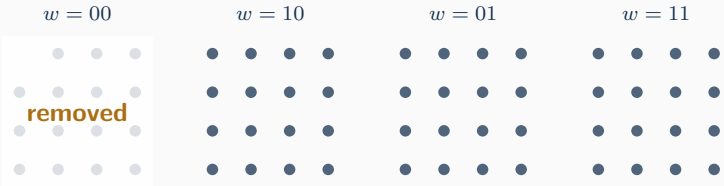
Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.



Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.

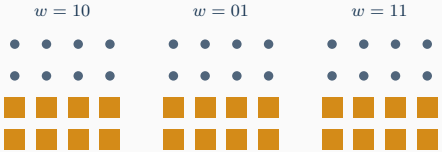


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ stabilizer code.

Hyperplane sections and codeword weights

A codeword of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



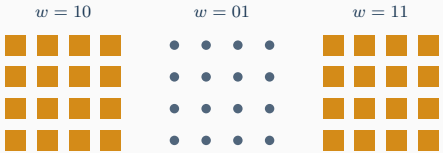
$\lambda \neq 0$, for example $\ell = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

Hyperplane sections and codeword weights

A codeword of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



$\lambda = 0$, for example $\ell = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p \backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no valid code.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0&0&0&0&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0&0&0&0&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

Q_{48}: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1x_2x_3\rangle = 1$, and $\langle \chi_G, m\rangle = 0$ for every other monomial of degree 1 to 3.

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1x_2x_3\rangle = 1$, and $\langle \chi_G, m\rangle = 0$ for every other monomial of degree 1 to 3.

Since $\mathrm{RM}(3, r)^\perp = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) is one solution [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

Logical rows in the column picture

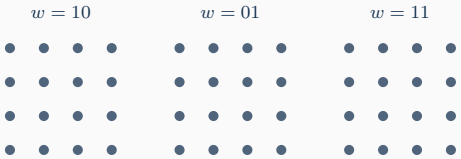
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



Logical rows in the column picture

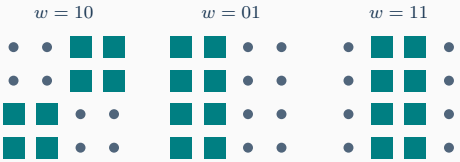
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 \oplus u_3$, $1 \oplus u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l \beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2, s))^{\perp}$, a linear condition.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l \beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2, s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, \beta_l \rangle_{\mathcal{P}} = 0$ (K2 and KS2).

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l\beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2,s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, \beta_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, \beta_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

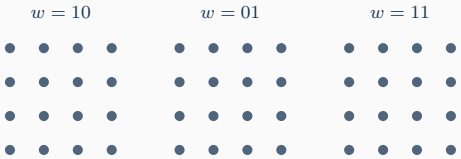
1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l\beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2,s))^{\perp}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, \beta_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, \beta_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

If no pair extends, $\mathcal{P}$ is excluded. A SAT solver can also solve the full system at once. For Q_{48}, the columns at $u = 0000$ in the three blocks sum to 0 and their labels to $011 \oplus 110 \oplus 000 = 101$. These three Z errors form a logical operator, so $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$



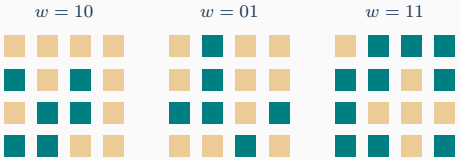
Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$

For Q_{48} [LB26]:

$$\begin{aligned}\gamma(u, w) &= u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u),\\ Q_1 &= u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4,\\ Q_2 &= u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.\end{aligned}$$



Teal: $T^\dagger$ $(6, 6, 10)$. Gold: T. Total $26\,T + 22\,T^\dagger$.

Stabilizer supports are Reed–Muller codewords

For projective S with support $\mathcal{P} \subset \mathbb{F}_2^s$, S1–S3 imply

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0 \pmod 2, \qquad 1 \le t \le 3.$$

Following [NH22], let $c = \chi_{\mathcal{P}} + \epsilon\delta_0$, where δ_0 is the indicator of the origin and ϵ makes $\mathrm{wt}(c)$ even. Every monomial of degree at least 1 vanishes at $\beta = 0$, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = \mathrm{wt}(c) \bmod 2$, which ϵ makes 0. Hence c is orthogonal to every monomial of degree at most 3, so $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3,s)^{\perp} = \mathrm{RM}(s-4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

For Q_{48}: $c = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no physical $T/T^\dagger$ layer [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat, punctured at the origin. Q_{48} removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with circuit footprint 9.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $Q_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Ours excludes $n \leq 38$, [WJS26] excludes 39–46.

Open questions

- Which supports admit compatible labels and signs?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public